

HORSE GPT

v 3 . 1 4

A Study in Probabilistic Form Analysis

DATA INGESTION

I.



BRIS DATA

1,430+ fields · fixed-width parse

II.



EQUIBASE

web scraper · entries & results

III.



DRF / API

morning line · historical payoffs

50 ENGINEERED FEATURES

z-scored within race field · 9 categories · relative strength

SPEED

(5)



PACE

(8)



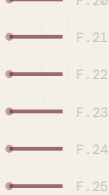
CLASS

(6)



FORM

(6)



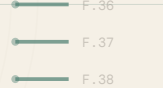
JOCKEY / TRAINER

(10)



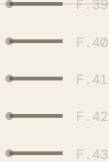
POST POSITION

(3)



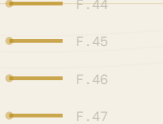
DISTANCE

(5)



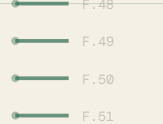
ODDS

(4)



EQUIPMENT

(4)



MODEL ENSEMBLE

LightGBM

w = 0.70

Benter Logistic

w = 0.30

log-odds space · softmax normalization

RACE PREDICTION

p = 1.0

Henery normal model · 100K Monte Carlo simulations · exotic bet pricing

Entry = Horse × Race · Market Odds as Logit Offset · Predictions Sum to 1.0

BRIS · EQUIBASE · DRF · 50 FEATURES · LIGHTGBM + BENTER · MONTE CARLO